

Bike Sharing Demand Forecasting

Business presentation for hourly rental planning

XGBoost

Selected model

50.91

Mean absolute deviation

0.886

R-squared

Prepared for planning and bicycle logistics stakeholders

Business Problem & Agenda

Focused discussion for a busy planning lead

- **Business question:**
how many bicycles should be ready by hour?
- Dataset overview and demand behavior
- **Exploratory data analysis:**
time, season, and weather effects
- Model performance comparison and XGBoost selection
- Operational recommendations and go-live view

XGBoost

Recommended model

50.91

Average miss per hour

R2 0.886

Model fit

Dataset Overview

Hourly rental demand with calendar and weather context

- **Source:**
UCI Bike Sharing Dataset, hourly Capital Bikeshare records.
- **Target:**
cnt, total rental count per hour.
- **Planning drivers:**
hour, weekday, working day, holiday, season, weather, temperature, humidity, and windspeed.
- **Leakage avoided:**
casual and registered are excluded because they are components of cnt.
- **Validation:**
chronological train/test split to mimic forecasting future demand.

17,379

Hourly records

cnt

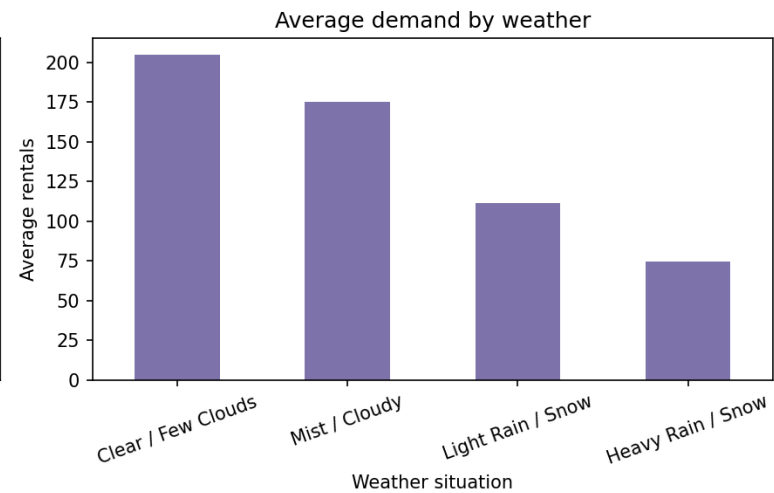
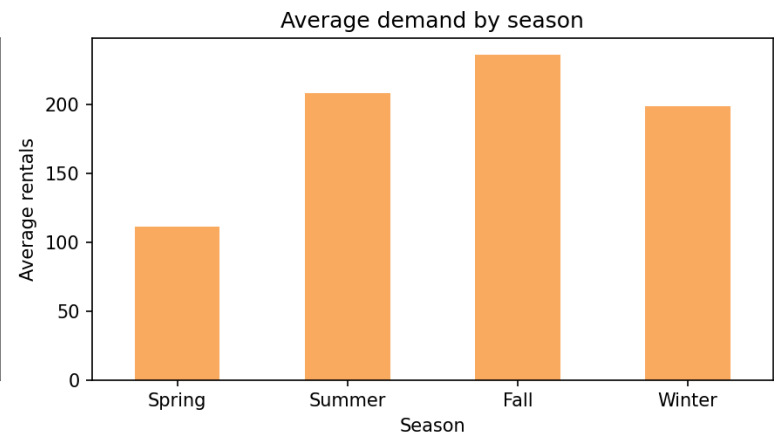
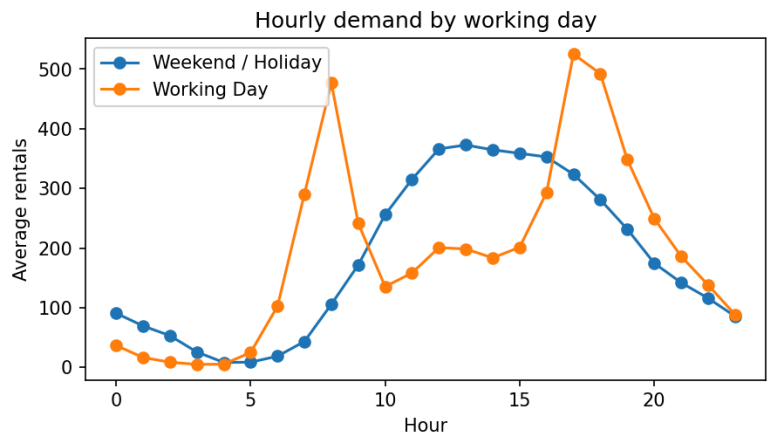
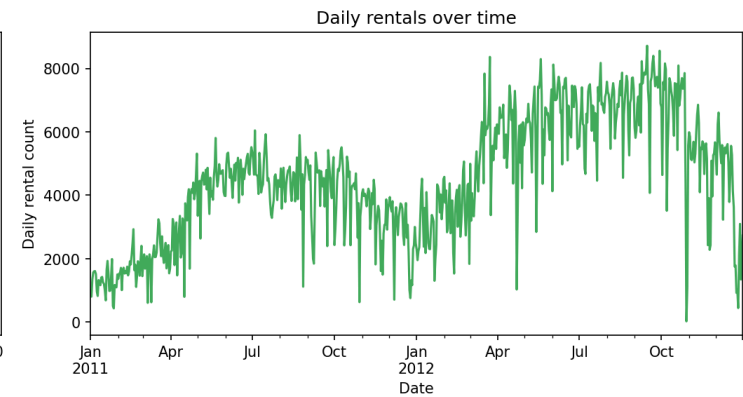
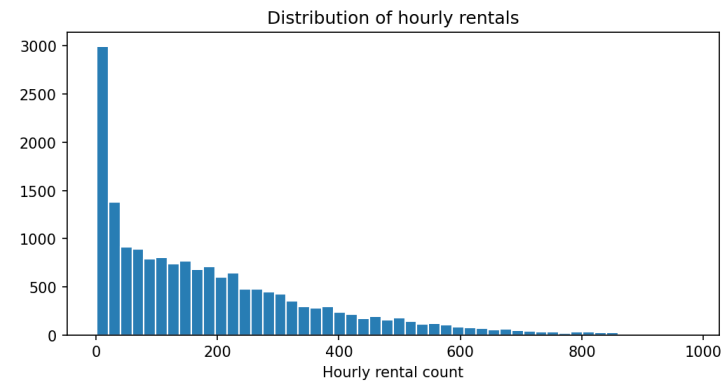
Target variable

Future holdout

Evaluation split

Exploratory Data Analysis

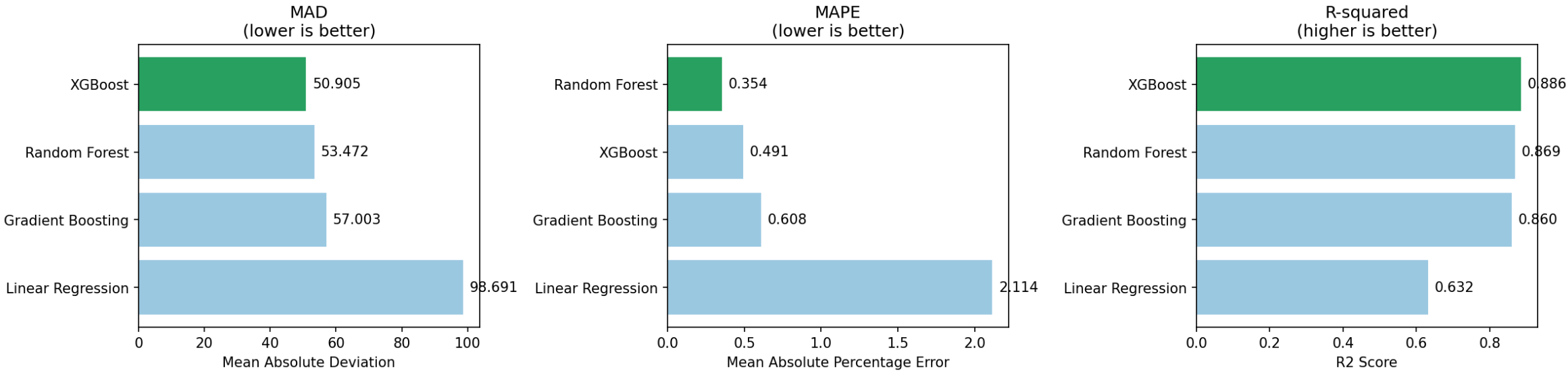
Demand is highly structured by time and season



Key message: peak demand is not random. Working-day commute hours, warm seasons, and clear weather require higher bicycle readiness.

Model Performance

All candidate models were compared on future holdout data

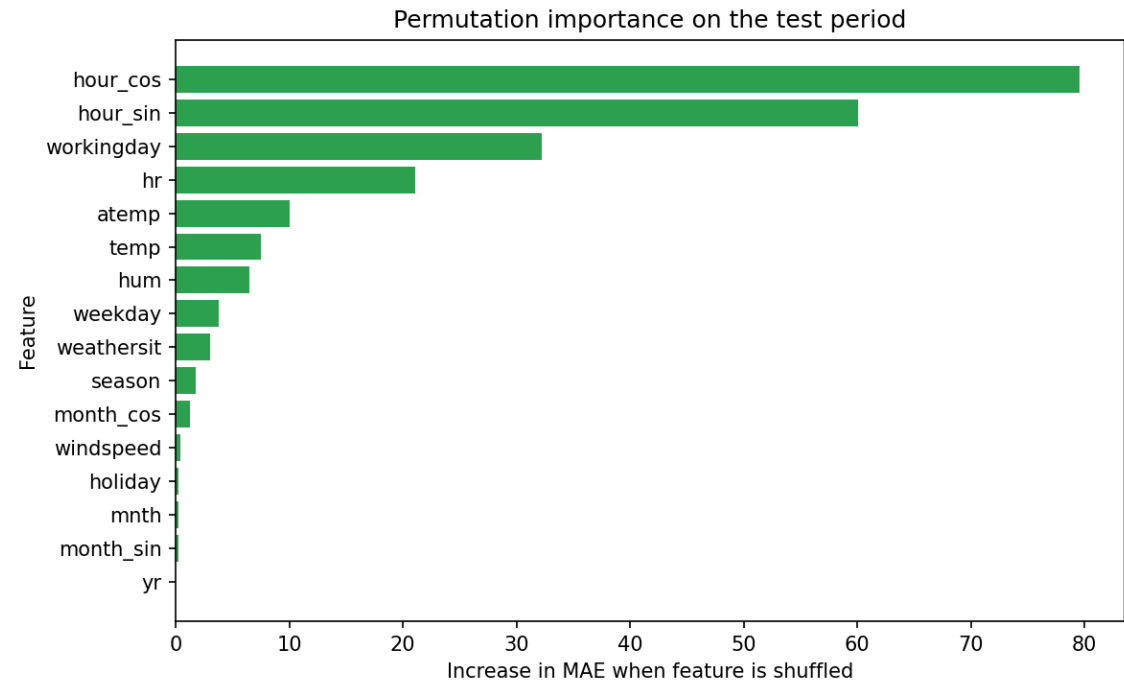


Decision: XGBoost gives the lowest mean absolute deviation and the strongest overall fit.

Final Model Selection

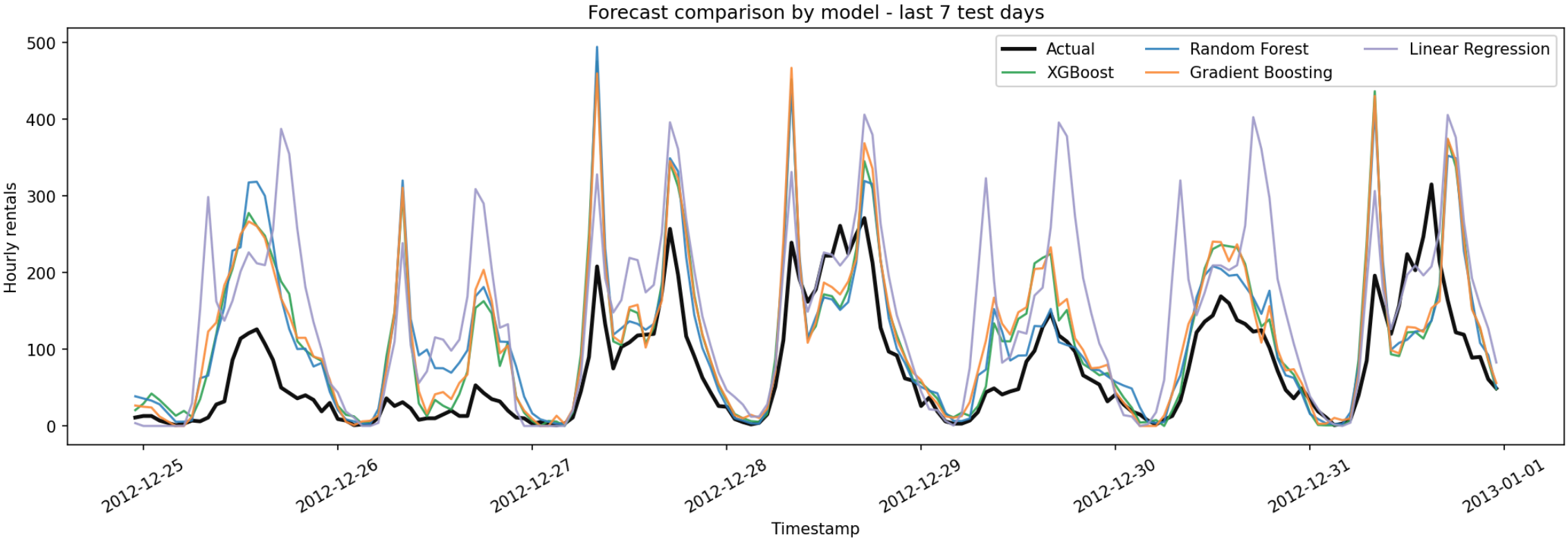
XGBoost is the best production candidate

- **Lowest MAD:**
50.91 hourly rentals.
- Captures nonlinear interactions between hour, working day, season, and weather.
- Practical to retrain and deploy as a single pipeline artifact.
- Supports an interactive Streamlit dashboard for scenario review.



Forecast Quality

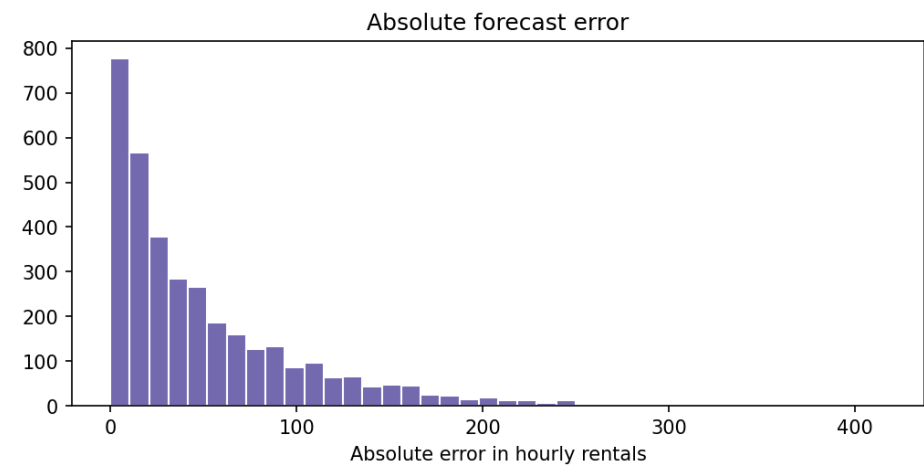
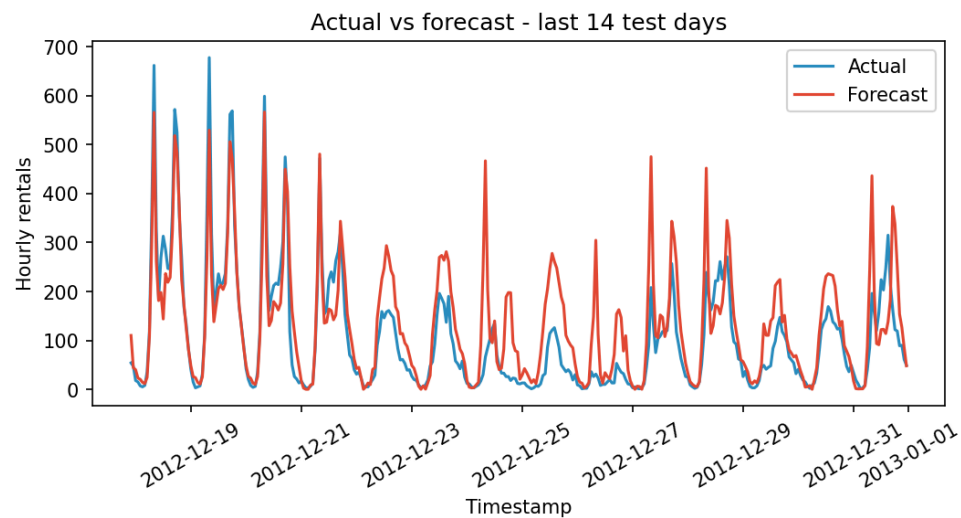
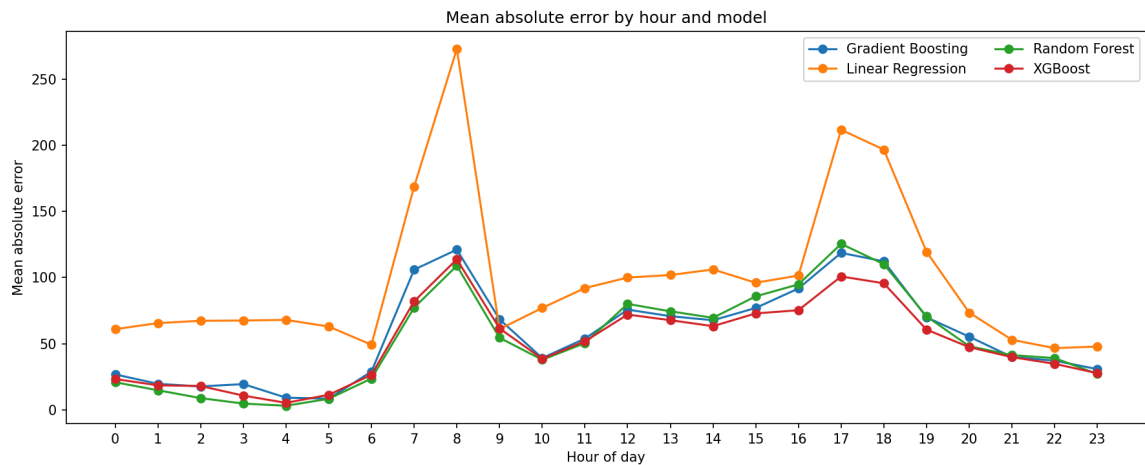
Model output supports an operational planning baseline



Use the forecast as a planning baseline, with operational buffers for peak periods and unusual local events.

Forecast Risk

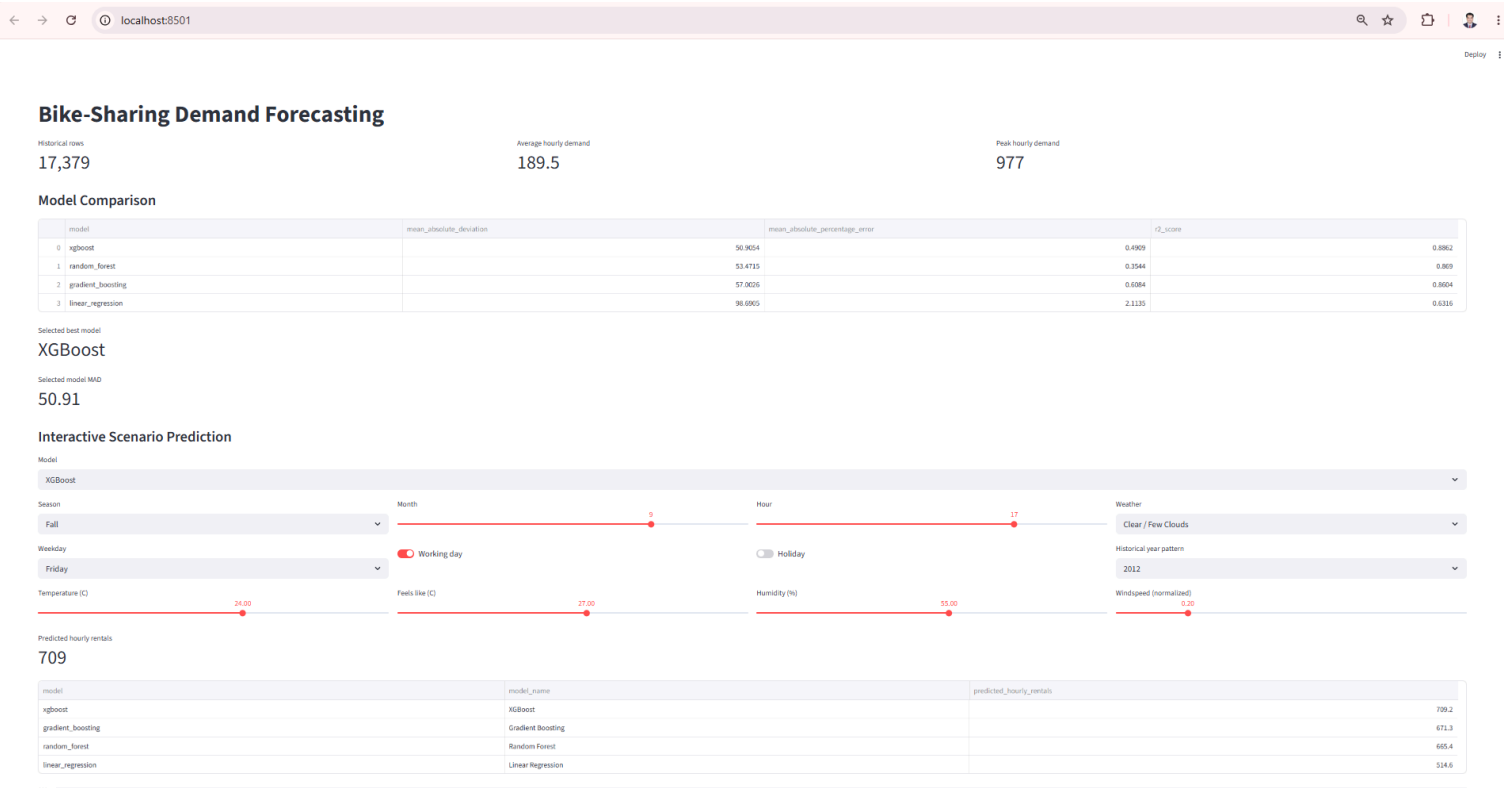
Error varies by hour and operating context



Operational Recommendations

How to use the model in planning

- Refresh forecasts daily after the latest completed demand data is available.
- Refresh intraday when weather changes materially or live demand deviates from plan.
- Use forecasts for short- to medium-term readiness, staffing, and repositioning.
- For annual planning, combine model output with fleet changes, station changes, pricing, holidays, and local events.



Summary: Quality Is Good For Go-Live

- XGBoost is the recommended model for the business case.
- Expected average hourly forecast miss: 50.91 rentals.
- The workflow is production-minded: validated data, reusable modules, saved artifacts, tests, reports, and dashboard.
- Go-live recommendation: start with daily forecasts plus monitoring by hour, season, and weather.

XGBoost

Selected model

50.91

MAD

Daily

Refresh